Comparative Analysis of Bandgap Reference Circuits Using Different Amplifier Topologies

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***Abstract*—For precise operation of analog and mixed signal ICs, a temperature independent reference voltage source is necessary. In this paper, we discuss the design of two types of band gap reference circuits using the diode. The circuits are designed and analyzed in a 180 nm CMOS technology using the CADENCE Virtuoso tool. Both designs use forward biased diodes to form CTAT and PTAT voltages, which are later added together to form a temperature independent reference voltage source.**

The main difference in design of the circuits is the type of feedback amplifier used; in one design, a two-stage NMOS differential amplifier is used, while in another design, a folded cascode amplifier is used. The performance of the designed circuits is studied through simulations under different operating conditions such as variations in temperature, changes in power supply voltage, and frequency response.

The findings show that the BGR circuit with folded cascode amplifier has better temperature independence, improved PSRR, and improved line and load regulation because of high gain and high output impedance. On the other hand, the differential amplifier-based BGR design shows low power consumption and simple circuit design.

***Index Terms*—Bandgap reference (BGR), CMOS analog circuits, diode-based reference, differential amplifier, folded cascode amplifier, temperature coefficient, PSRR, line regulation.**

I. Introduction

Voltage reference circuits are one of the essential components in the design of analog and mixed-signal integrated circuits. Analog-to-digital converters (ADCs), digital-to-analog converters (DACs), power management units, phase-locked loops (PLLs), sensor interfaces, and System-on-Chip (SoC) subsystems all need a reference voltage that is stable and accurate. The accuracy of the reference has a direct impact on the system's linearity, noise margin, and overall performance. A voltage reference should not change when the temperature, supply voltage, process changes, or load conditions change.

The bandgap reference (BGR) circuit, which Widlar first came up with in 1971 and Brokaw and Kuijk later improved, is now the standard way to make reference voltages that don't change with temperature in both CMOS and bipolar technologies. The basic idea behind the BGR is to cancel out two temperature-dependent quantities that have opposite properties:

- **CTAT (Complementary-to-Absolute-Temperature):** The base–emitter voltage (V_{BE}) of a forward-biased p–n junction decreases with temperature at approximately $-2\,\mathrm{mV/^\circ C}$.
- **PTAT (Proportional-to-Absolute-Temperature):** The thermal voltage ($V_T = kT/q$) increases linearly with absolute temperature at approximately $+0.085\,\mathrm{mV/^\circ C}$.

When these two parts are properly scaled and added together, the voltage comes out to be about $1.205\,\mathrm{eV}$ at 0 K, which is close to the extrapolated silicon bandgap energy. This gives a reference voltage of about $1.2\,\mathrm{V}$ with a first-order temperature coefficient that is almost zero.

Parasitic bipolar transistors, diode-connected MOSFETs, or forward-biased p-n junction diodes can be used to make CTAT and PTAT parts in CMOS technology. The diode-based method is especially good for standard digital CMOS processes that don't make high-quality bipolar devices because it uses the built-in source-bulk or drain-bulk junctions of MOSFETs.

The feedback operational amplifier in a BGR circuit ensures accurate summation of PTAT and CTAT components by enforcing equality conditions at the summing node, thereby minimizing offset-induced temperature drift.

The type of feedback amplifier you choose has a big effect on important performance factors like temperature coefficient (TC), power supply rejection ratio (PSRR), line regulation, and noise. The standard two-stage NMOS differential amplifier has a moderate open-loop gain (usually 60–80 dB in 180 nm CMOS), is easy to design, and can easily be compensated for frequency. But its low gain and PSRR can make reference accuracy worse when the temperature and supply voltage change. The folded cascode operational amplifier, on the other hand, has a higher open-loop gain (usually between 80 and $100\,\mathrm{dB}$) and a better PSRR because it has a high output impedance. However, this makes the circuit more complicated and uses more power.

This paper presents the design, simulation, and comparative analysis of two BGR circuits implemented in the GPDK 180 nm CMOS process using Cadence Virtuoso: (1) a BGR employing a two-stage NMOS differential amplifier, and (2) a BGR utilizing a folded cascode amplifier. In both designs, forward-biased diodes are used to generate PTAT and CTAT components. The temperature coefficient (TC), PSRR, line

regulation, load regulation, and transient response are used to compare the two topologies.

II. Literature Review

A. Evolution of Temperature Compensation Techniques

The bandgap reference (BGR) concept is based on combining two voltages with opposite temperature coefficients to achieve a temperature-independent reference. The base–emitter voltage (VBE) of a bipolar junction transistor, or the forward voltage (VF) of a diode, exhibits a Complementaryto-Absolute Temperature (CTAT) characteristic, decreasing approximately linearly with temperature at a rate close to 2 mV/∘C.

In contrast, a Proportional-to-Absolute Temperature (PTAT) voltage is generated from the difference in voltages of two junctions operating at different current densities. By properly scaling and summing the CTAT and PTAT components, a reference voltage with minimal temperature dependence can be realized.

In standard 180 nm CMOS technology, first-order compensation techniques usually achieve temperature coefficients (TC) between 20 and 50 ppm/°C. Modern precision systems require TC values below 10 ppm/°C.

B. Feedback Topologies: Differential and Folded Cascode

The error amplifier becomes an integral part of BGR circuits as the means of achieving balance between PTAT and CTAT branches.

- Two-Stage NMOS Differential Amplifier: The classical two-stage NMOS differential amplifier is popular due to its easy-to-design nature and rather high output swing. However, it has mediocre gain (60-70 dB in 180 nm CMOS technology) and poor PSRR, which causes offset errors and negatively affects temperature stability.
- Folded Cascode Amplifier: The folded cascode approach allows for obtaining high gain (over 80 dB) and output resistance within one stage and thus provides for small output errors and a good reference voltage generation.

Additionally, the folded cascode amplifiers provide for better rejection of power supply noise, thus being more suitable for modern ICs. The drawback of the folded cascode amplifiers is their complicated design process and low output swing.

C. Diode-Based Reference Generation in Standard CMOS

Present-day CMOS bandgap references make extensive use of diode connected devices to create the required CTAT nature of the voltage since it can be fabricated with common processes and without using special bipolar devices. Diode type circuits have an advantage of small physical footprint and low-voltage capability that allows their use in highly integrated mixed signal circuits. The properties of the reference circuit related to its temperature dependence are determined by diode forward voltages and the precision of the PTAT current source. Therefore, the accuracy of the error amplifier is an important issue.

D. Recent Circuit Innovations in BGR Design

Recent research has focused on pushing the performance limits of BGR circuits in terms of temperature stability, supply rejection, and low-voltage operation. Advanced feedback and regulation loop techniques have demonstrated simultaneous improvement in PSRR and output stability. High-order temperature compensation in 0.18 µm CMOS has achieved sub-1 ppm/°C temperature coefficients, establishing that first-order cancellation alone is insufficient for precision applications. Reviews of high-accuracy BGR topologies suggest that cascoded amplifier structures and curvature correction are the primary enablers of performance below 20 ppm/°C.

On the low-voltage front, references operating at 0.5 V with temperature coefficients around 16 ppm/°C have been reported, reflecting the growing demand for sub-1 V designs in IoT and wearable applications. Comparative studies of cascode, two-stage, and telescopic opamp architectures consistently confirm that folded cascode configurations yield the highest gain and output impedance at the cost of increased design complexity.

E. Summary of Technical Gap

A lot of research has been done on bandgap reference design but not many studies directly compare the standard NMOS differential amplifier with the folded cascode amplifier inside a diode-based bandgap circuit using 180 nm GPDK technology. This work fills that gap by comparing both designs in terms of temperature stability, power supply rejection, and line regulation using Cadence Virtuoso simulations.

III. Methodology

The proposed bandgap reference (BGR) design is based on the principle of generating a temperature-independent reference voltage by combining two complementary temperature-dependent voltages: Complementary-to-Absolute Temperature (CTAT) and Proportional-to-Absolute Temperature (PTAT). The CTAT voltage is derived from the forward voltage of a diode, which decreases with temperature, while the PTAT voltage is obtained from the difference between two diode voltages operating at different current densities.

A. Circuit Topology Selection

Two amplifier topologies are used in this work to investigate their impact on BGR performance:

- **Two-Stage NMOS Differential Amplifier:** A simple and widely used architecture offering ease of design and wider output swing, but with moderate gain and poor PSRR performance.
- **Folded Cascode Amplifier:** A system with high gain that offers better output impedance and PSRR but increased design and power requirements.

The main aim in utilizing these two circuits is to study their impact on various performance metrics like temperature stability and efficiency.

B. Design Implementation

The BGR circuits were implemented using GPDK 180 nm CMOS technology in Cadence Virtuoso. Both architectures employ diode-connected devices to generate CTAT and PTAT voltages. A resistor network is used to appropriately scale and combine these components to generate a stable reference voltage (V_{ref}).

TABLE I
DESIGN PARAMETERS

Parameter	Value
Supply Voltage V_{DD}	1.8 V
W/L (BGR)	2 µm / 180 nm
W/L (Folded Cascode OTA)	1.08 µm / 180 nm
R_6 (Differential BGR)	8.89 kΩ
R_7 (Differential BGR)	1 kΩ
R_6 (Folded Cascode BGR)	4.445 kΩ
R_7 (Folded Cascode BGR)	1 kΩ
Diode Ratio n	8

The operational amplifier in each topology enforces feedback, ensuring proper balance between PTAT and CTAT branches and maintaining a stable operating point.

Bias currents are generated using current mirror circuits, and transistor sizing is optimized to ensure saturation operation, adequate gain, and reduced mismatch effects.

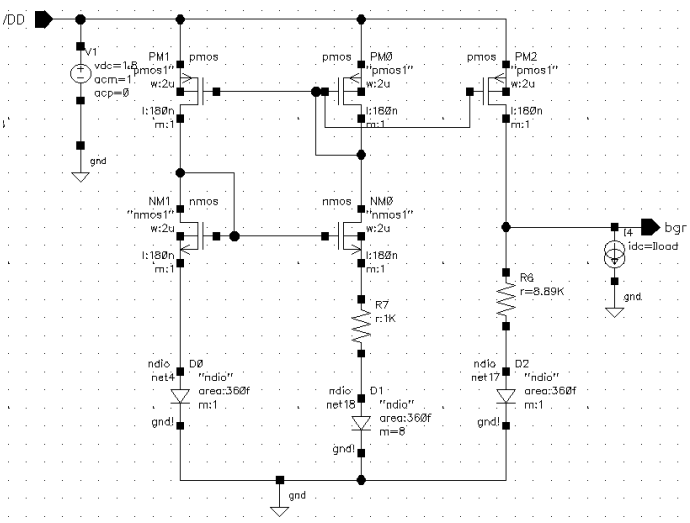


Fig. 1. Bandgap reference circuit using two-stage NMOS differential amplifier.

C. Simulation Strategy

To evaluate performance, both designs were simulated under identical operating conditions using Spectre simulator. The following analyses were performed:

- **Temperature Sweep:** $-50°$C to $125°$C to evaluate temperature stability of V_{ref}.
- **Line Regulation:** Supply voltage variation from 1.8 V to 3.3 V to study sensitivity to supply changes.
- **PSRR Analysis:** AC analysis with small-signal supply perturbation to evaluate noise rejection capability.

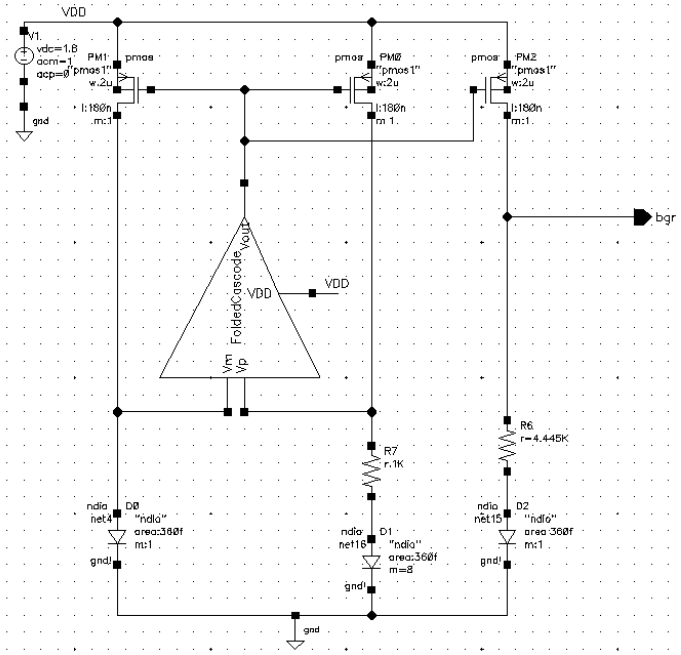


Fig. 2. Bandgap reference circuit using folded cascode amplifier.

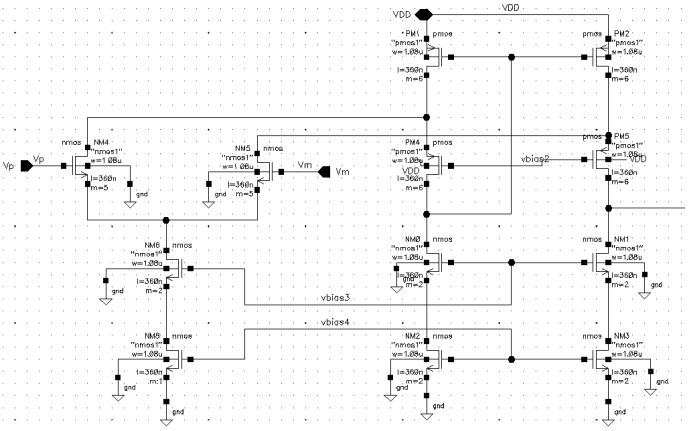


Fig. 3. Folded cascode operational amplifier used in the BGR design.

- **Load Regulation:** Variation of load current to analyze output voltage stability under different loading conditions.
- **Power Consumption:** Measured using DC operating point analysis.

Power consumption is calculated as:

$$P = V_{DD} \times I_{DD}$$

D. Performance Evaluation Metrics

The performance of both BGR architectures is evaluated using the following key metrics:

- Temperature Coefficient (TC)
- Line Regulation
- Power Consumption

- Power Supply Rejection Ratio (PSRR)
- Load Regulation

These parameters are extracted from simulation results and used to perform a comparative analysis of both amplifier topologies in terms of stability, accuracy, and power efficiency.

IV. Results and Discussion

The performance of the two bandgap reference (BGR) topologies—two-stage NMOS differential amplifier and folded cascode amplifier—is evaluated using Cadence Virtuoso simulations. For each performance metric, both designs are analyzed separately and compared.

A. Temperature Analysis

The temperature variation of the reference voltage (V_{ref}) was analyzed from $-50°$C to $125°$C.

$$TC = \frac{1}{V_{\text{ref}}} \cdot \frac{dV_{\text{ref}}}{dT}$$

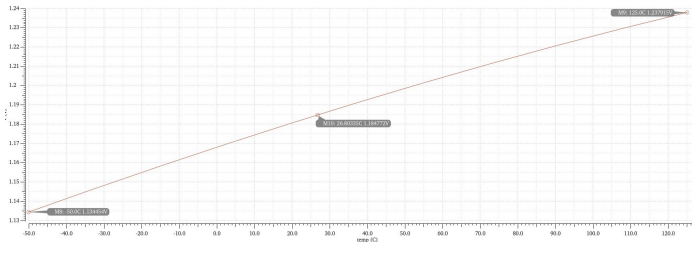


Fig. 4. Temperature variation of V_{ref} for differential amplifier-based BGR.

$$TC \approx 498\,\text{ppm}/°C$$

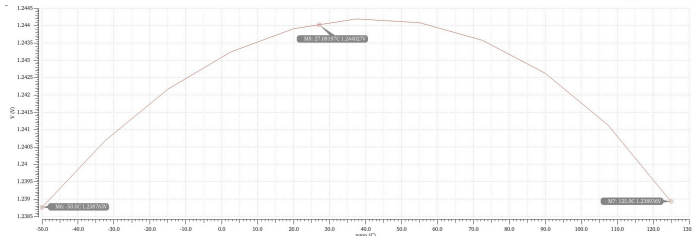


Fig. 5. Temperature variation of V_{ref} for folded cascode-based BGR.

$$TC \approx 0.80\,\text{ppm}/°C$$

The folded cascode-based BGR shows significantly improved temperature stability with a flatter response over the entire range. This is due to higher loop gain, which improves CTAT and PTAT cancellation accuracy. The differential amplifier exhibits higher variation due to limited gain and offset effects.

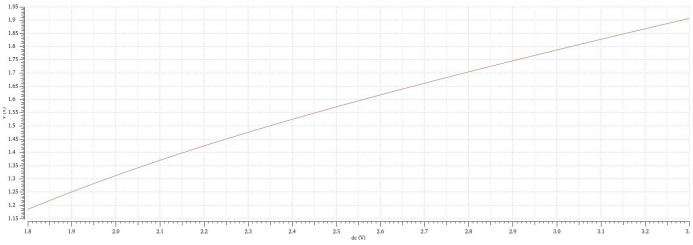


Fig. 6. Line regulation for differential amplifier-based BGR.

B. Line Regulation Analysis

The supply voltage was swept from 1.8 V to 3.3 V.

$$\text{Line Regulation} = \frac{\Delta V_{ref}}{\Delta V_{DD}}$$

$$\text{Line Regulation} \approx 490\,\text{mV/V, of Differential Pair}$$

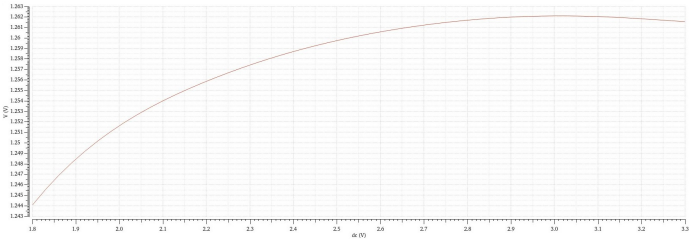


Fig. 7. Line regulation for folded cascode-based BGR.

$$\text{Line Regulation} \approx 11.7\,\text{mV/V, of Folded Cascode}$$

The folded cascode architecture demonstrates better line regulation due to higher output impedance and improved supply rejection. The differential design shows noticeable variation with supply voltage changes.

C. Power Consumption

The power consumption was calculated from the DC operating point by measuring the supply current (I_{DD}):

$$P = V_{DD} \times I_{DD}$$

$$\text{Power Consumption} \approx 149.877\,\text{uW, of Differential Pair}$$

$$\text{Power Consumption} \approx 660.117\,\text{uW, of Folded Cascode}$$

The differential amplifier-based BGR consumes less power due to its simpler structure and lower bias current requirements. The folded cascode design, while offering better performance, consumes higher power due to additional biasing branches and increased circuit complexity.

D. Power Supply Rejection Ratio (PSRR)

AC analysis was performed to evaluate supply noise rejection.

The folded cascode design achieves better PSRR performance, especially at low frequencies, due to cascoding which isolates the output from supply noise. The differential amplifier shows lower rejection due to limited gain.

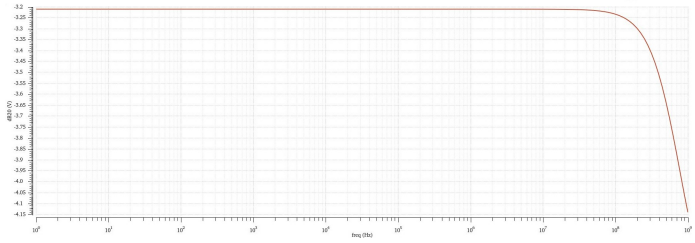


Fig. 8. PSRR vs frequency for differential amplifier-based BGR.

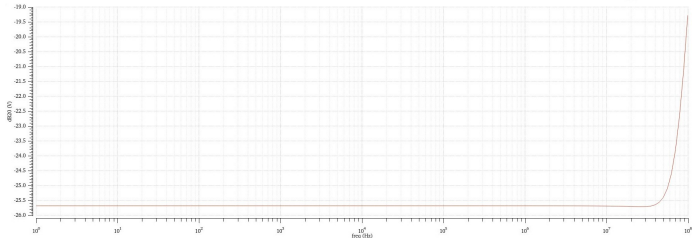


Fig. 9. PSRR vs frequency for folded cascode-based BGR.

E. Load Regulation Analysis

The load current was varied to observe output stability.The folded cascode topology maintains a more stable output under varying load conditions due to higher output resistance. The differential design shows slightly higher variation in V_{ref}. The folded cascode has 6× lower output resistance = much better load regulation.

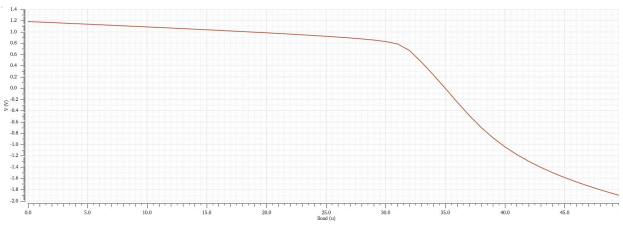


Fig. 10. Load regulation for differential amplifier-based BGR.

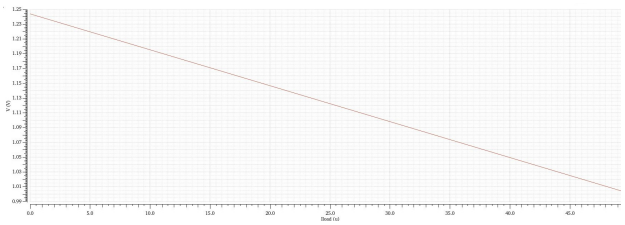


Fig. 11. Load regulation for folded cascode-based BGR.

F. Performance Comparison

Across all performance metrics, the folded cascode amplifier-based BGR consistently outperforms the differential amplifier-based design in terms of temperature stability, PSRR, line regulation, and load regulation. However, the differential amplifier offers lower power consumption and simpler implementation, making it suitable for low-power applications where ultra-high precision is not critical.

A comparative summary of the two designs is presented in Table II.

TABLE II
PERFORMANCE COMPARISON OF BGR DESIGNS

Parameter	Differential	Folded Cascode
Temperature Coefficient	498ppm/$^\circ C$	0.80ppm/$^\circ C$
PSRR	-3.2 dB	-26 dB
Line Regulation	490 mV/V	11.7 mV/V
Power Consumption	149.877 uW	660.117 uW
Load Regulation	30 kΩ	5.3kΩ

V. CONCLUSION

In the current research, two circuits of diode-based bandgap reference (BGR) with various amplifiers used have been designed and examined in a 180 nm CMOS process. Based on the simulation results, the type of feedback amplifier used has an important effect on the characteristics of the generated reference voltage in terms of its accuracy, stability, and robustness. Thus, the temperature coefficient of the reference voltage generated in the folded cascode circuit equals 0.80 ppm/°C while that of the voltage generated using differential amplifier equals 498 ppm/°C. This difference means that folded cascode architecture provides better resistance to temperature changes.

The lower PSRR of 26 dB and better line regulation of 11.7 mV/V in the folded cascode-based circuit mean that the latter provides much higher resistance to voltage fluctuations and noise in comparison with the first one. Consequently, the reference voltage remains stable and provides enhanced performance of the circuits where it is used, such as ADCs, DACs, regulators, and sensor interface circuits. The ability of the folded cascode circuit to keep the regulation even at a relatively low load resistance of 5.3 k shows that the architecture has better load driving capability.

This is achieved at the cost of increased power consumption since the folded cascode consumes 660.117 µW against the 149.877 µW consumed by the differential amplifier BGR circuit. Thus, although the differential amplifier topology appears to be an ideal choice for low-power operations in which energy efficiency becomes the most important factor, the relatively high temperature drift, poor power supply rejection, and weak regulation capability make it unsuitable for generating accurate references. In summary, the folded cascode amplifier presents a better choice for high-performance BGR circuits.

VI. Future Scope

The paper made a comparison between the performances of diode-based BGRs that use differential and folded cascode amplifiers in an 180 nm CMOS process. In order to compare the intrinsic features of the two amplifiers, both architectures have been modeled as base cases without adding any other enhancements like improving PSRR, curvature correction, trimming, or offset cancellation. Consequently, the differences found in terms of temperature stability, supply rejection, and regulation can be attributed to the amplifier architecture.

Future investigations could extend the comparison by doing post-layout simulation in order to analyze how parasitic effects, mismatches, and layout variations influence both architectures. The robustness of the architectures could also be studied by performing process corners analysis and Monte-Carlo analysis to investigate how sensitive they are to process variation. Further optimization of transistors' sizes and bias conditions can be done for both architectures especially since the folded cascode architecture is accurate but power consuming.

Additionally, the comparative study can be expanded to include other amplifier topologies, such as telescopic cascode, gain-boosted cascode, and multistage amplifiers, thereby providing a broader understanding of how amplifier architecture influences the overall performance of CMOS bandgap reference circuits.

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